KHUSHAL SINGH

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EDUCATION

Carleton University **Ottawa, ON**

BCS Honours, Computer and Internet Security Stream, minor in Mathematics

- Major GPA: 9.81/12, Dean's Honour List 2022

TECHNICAL SKILLS

Languages: C++, C, Java, Python, Objective-C, C#, JavaScript, XML, ReactJS, Shell Scripting, Genero
Technologies: Jenkins, Linux, Perforce, GitHub, TeamCity, JIRA, Docker, QT framework, Postman, Confluence
Databases: MongoDB, MySQL, NoSQL, SQL, PostgreSQL

EMPLOYMENT

Engineer II **Versaterm Public Safety** **Jan 2025 – Present**

- Engineered and processed semantically structured NG911 (Next Generation 911) payloads by refining components such as location data, emergency situations, agencies, and agents within the Computer-Aided Dispatch (CAD) application.
- Led backend software upgrades and feature implementations to support new client contracts for Police, Fire, and EMS departments across the United States and Canada.
- Integrated third-party applications with the CAD system and resolved interoperability issues to enhance emergency services support, enabling reliable transmission and reception of incident data via platforms such as Motorola, Solacom, Micro Automation, Frequentis, APCO, ProQA, and Corti triage systems.
- Resolved defects and improved functionality in the CAD-8 application using Genero and C#, ensuring seamless integration with Mobile Data Terminal (MDT) applications used by police units in vehicle

Software Engineer **Motorola Solutions** **Jan 2021 – Oct 2022**

- Worked on MDS configurator for VESTA 911 Emergency Call Handling Software to develop an algorithm that reduced data validation time from 50 minutes to 6 minutes (83% enhancement)
- Efficiently inserted, extracted, and modified data from a PostgreSQL database using Python
- Developed test cases using C# and XML to validate the system before version upgrade deployments
- Resolved 25+ security vulnerabilities during infrastructure health checks on different labs
- Used Angular.ts and Material library to transform MDS user interface, reducing load time by 80% and improving user friendliness
- Implemented automated white-source scans within testing pipelines across 10+ projects on TeamCity platform to ensure compliance of software dependencies with US Federal Laws
- Took charge of Scrum meetings in the Scrum Master's absence, ensuring seamless communication and progress towards sprint goals, and presented updates during biweekly sprint / town hall meetings

Software Infrastructure Developer Co-Op **Ford Canada** **Sept 2020 – Dec 2020**

- Collaborated with ECG team consisting of 26 developers to update user profiles on BlackBerry's QNX infotainment software within Ford vehicles
- Created strategic unit test (UUT) cases for software functionality using C++, leveraging Google Test and Google Mock frameworks to achieve 100% code coverage
- Documented designs, validated efficacy, and utilized Jenkins to address bugs in 10+ generated software builds
- Led agile C++ Development, ensuring alignment with project goals and facilitating iterative advancement

Web Developer Co-Op **Carleton University** **June 2020 – Aug 2020**

- Worked on the sWall project – a large interactive wall of screens showcasing the history, present activities, and success stories of the SCE Department to 4000 students
- Wireframed the system through Balsamiq sketches and developed it using Vue.js and CSS Bootstrap
- Deployed Web-CAT, an automated assignment grading tool, within Docker by integrating Apache Tomcat Server with Java Version 8 and a MySQL database
- Published textual and video documentation showing the process
- Built a Self-Assessment portal (CRUD app) where 4th-year engineering uploaded and received feedback on their capstone projects
- Launched a PyMongo-Flask application, using React.js for frontend and Python for backend, with MongoDB as the database; managed the exchange of data between frontend and backend using Axios API